

Smart Bahrain Traffic Application





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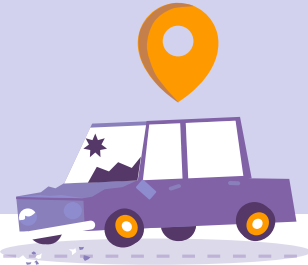
**Requirement
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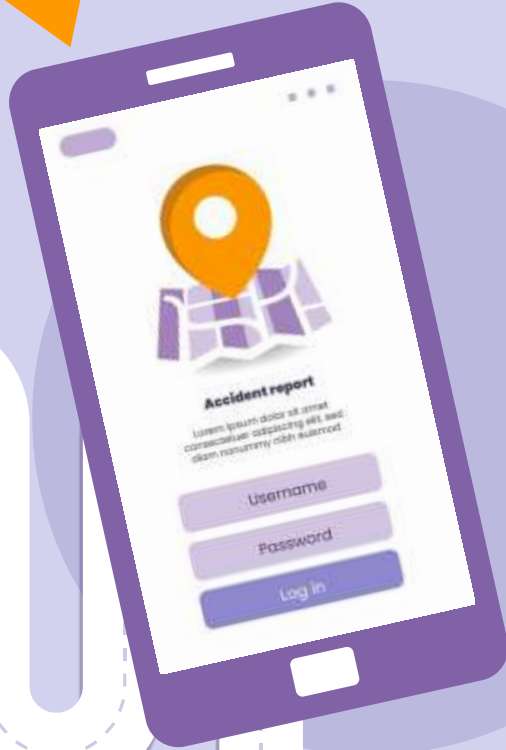
**System
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User interface

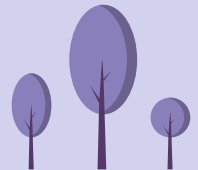


ACCIDENT!



Introduction

- Traffic and parking violations reduce road safety and traffic efficiency
- Many violations go unreported due to limited enforcement resources
- Increasing vehicle numbers require smart, community-based solutions



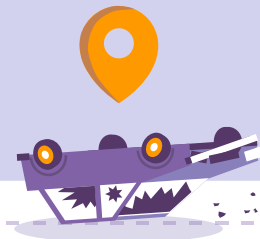
Problem Definition

- Traffic and parking violations reduce road safety and cause congestion
- Limited manpower makes it difficult to monitor all areas
- Manual reporting is slow and inefficient
- Many violations go unreported, weakening traffic management



Project aims

- Develop a mobile app for quick reporting of traffic and parking violations
- Improve communication between citizens and traffic authorities
- Enable faster and more accurate handling of reported incidents
- Support digital transformation and safer, organized roads



Research finding

Citizen reporting systems

- Citizen reporting apps enable the public to report violations using photos or videos
- Common features include location tagging, evidence upload and centralized authority review
- Such systems improve community participation in public safety



Targeted users

Targeted Customers or Beneficiaries

- Citizens and Residents: Report local traffic and parking violations.
- Traffic Authorities: Review reports, verify evidence and take action.
- Policy Makers: Use data to identify issues and improve regulations.
- Organizations: Promote road safety and responsible driving.





Requirement Elicitation

Requirements identified through:

- Personal observations of traffic and parking issues in Bahrain.
- User experiences shared by residents and friends.
- Analysis of existing systems.
- Literature review of similar apps (Traffic Prahari in India, Saudi Balady)

highlighted key features:

- Evidence upload (images/videos)
- Report tracking
- Efficient authority review



Functional Requirements

- User registration and secure login
- Identity verification (ID, email, or phone)
- Submit violation reports with details
- Upload photo and video evidence
- Automatic GPS location capture
- Authority review and report verification
- Real-time report status tracking
- Status change notifications
- Secure storage of reports and evidence
- Rewards for confirmed reports



Non Functional Requirements

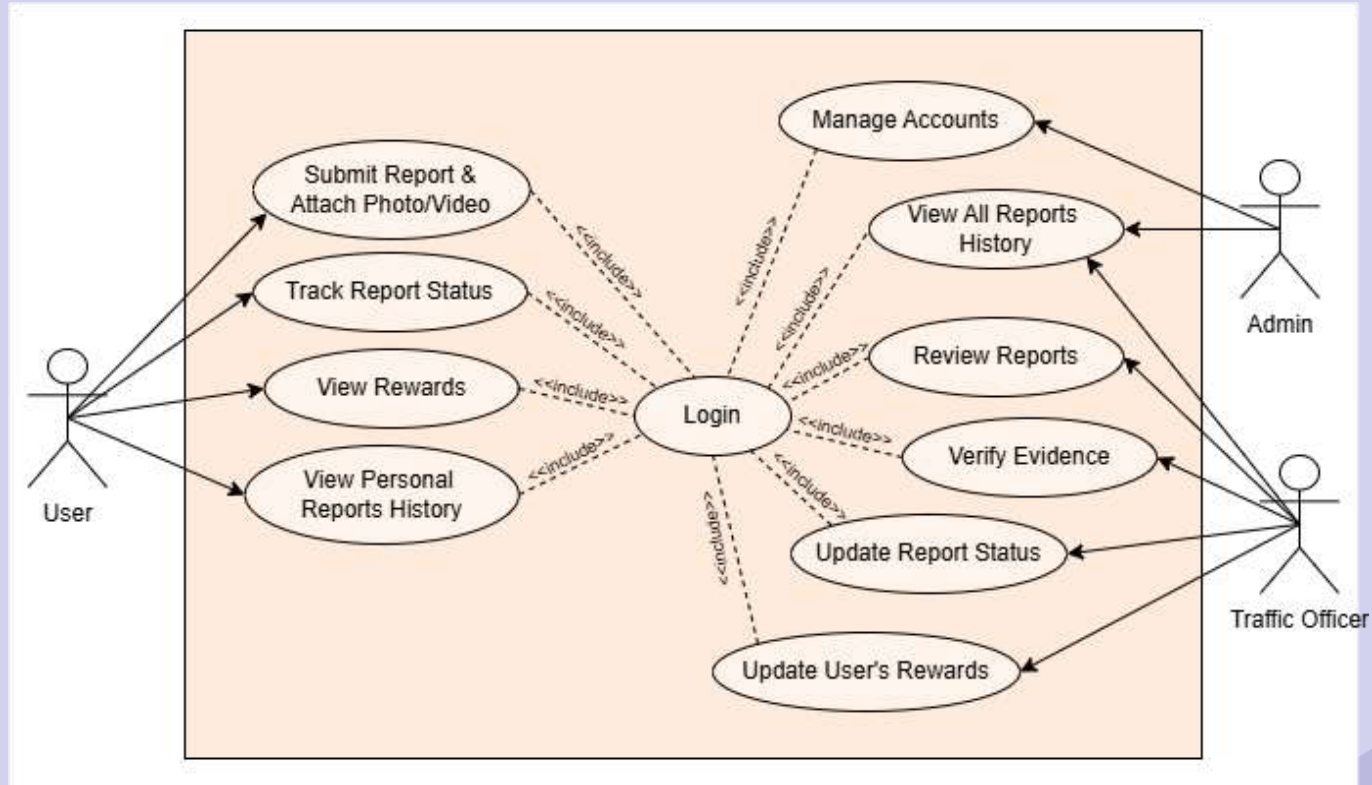
- **Performance:** Fast loading and efficient report handling
- **Security:** Encrypted data and controlled access
- **Usability:** Simple, intuitive interface for all users
- **Reliability:** Stable operation during peak usage
- **Scalability:** Supports future features and violation types
- **Maintainability:** Modular code for easy updates

Software Development Model

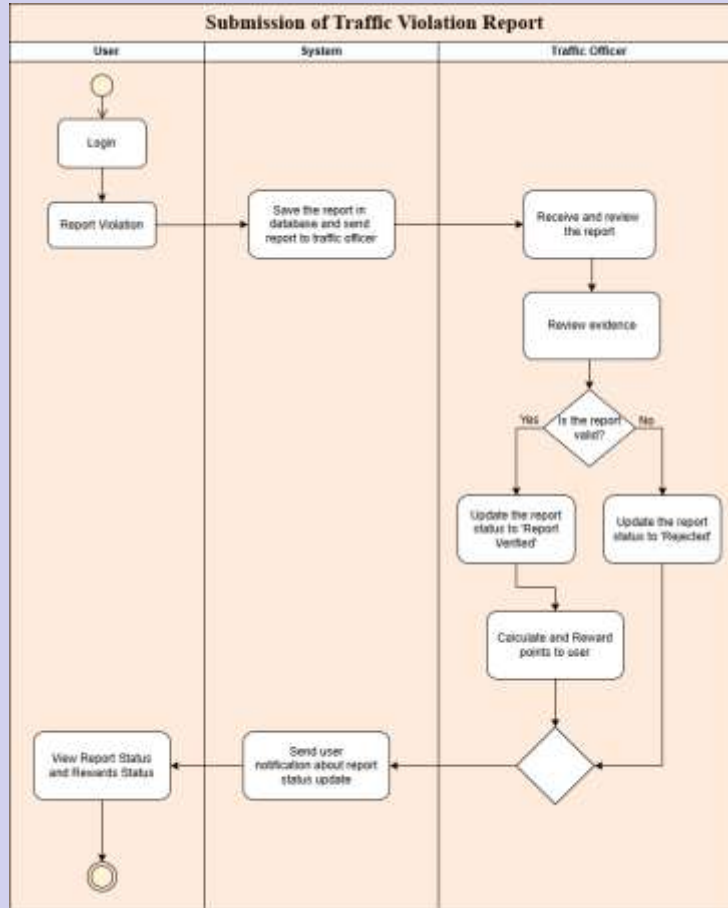
Model Used: Incremental Development Model

- System developed in small, independent modules
- Core features delivered and tested early
- Allows continuous feedback and improvements
- Reduces risk through early testing and modular development
- Supports scalability, flexibility and easy maintenance

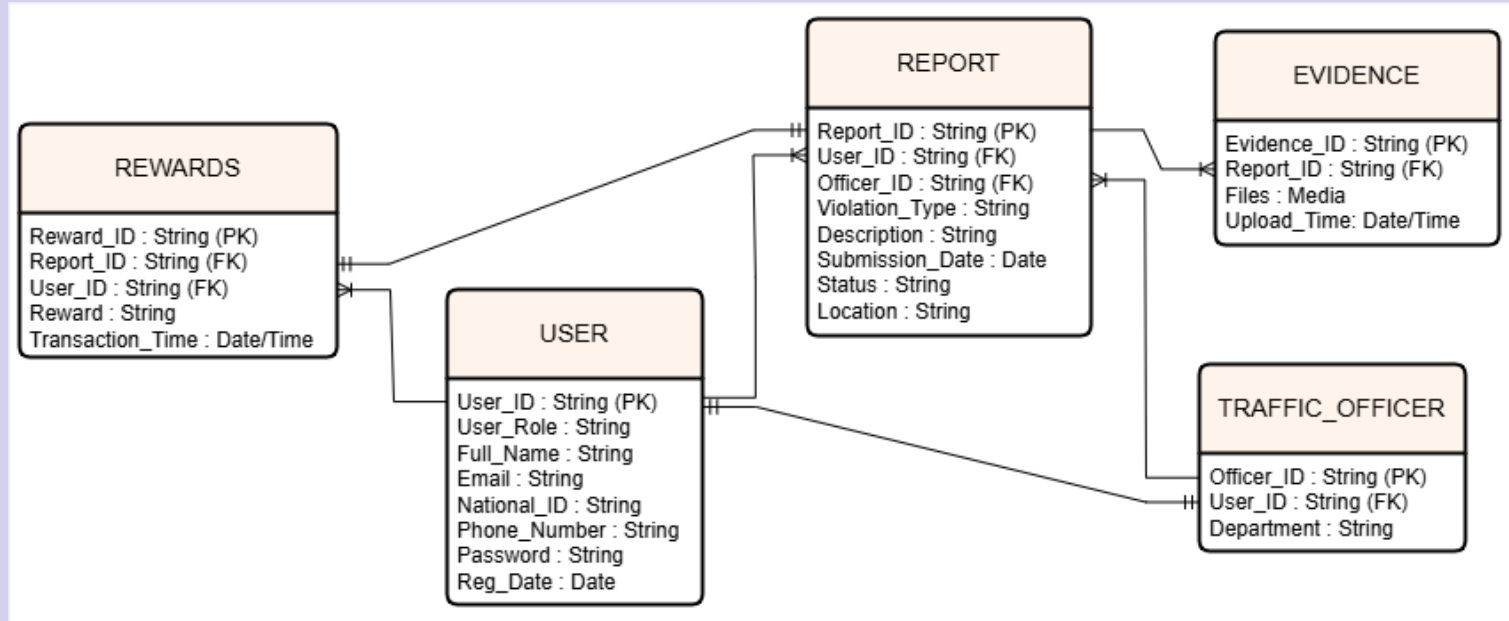
Use-Case Diagram



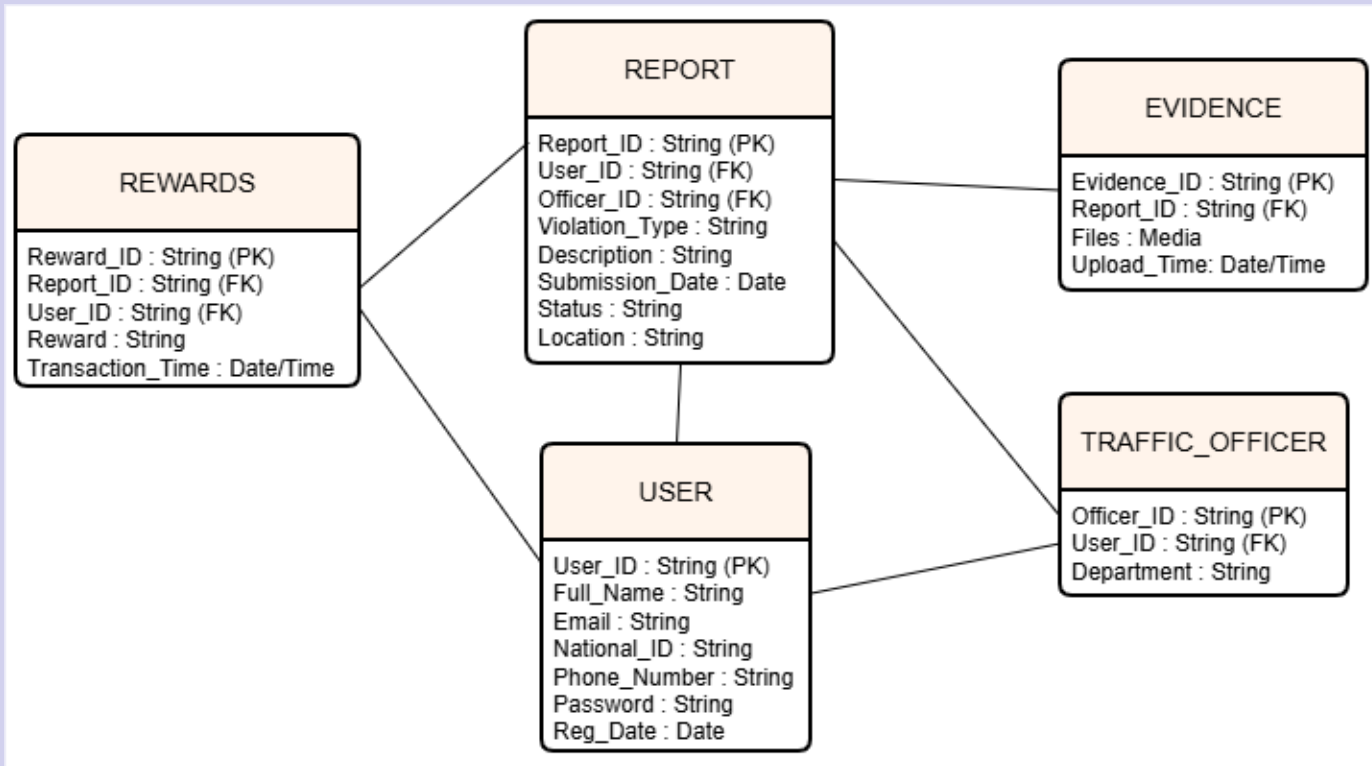
Activity Diagram



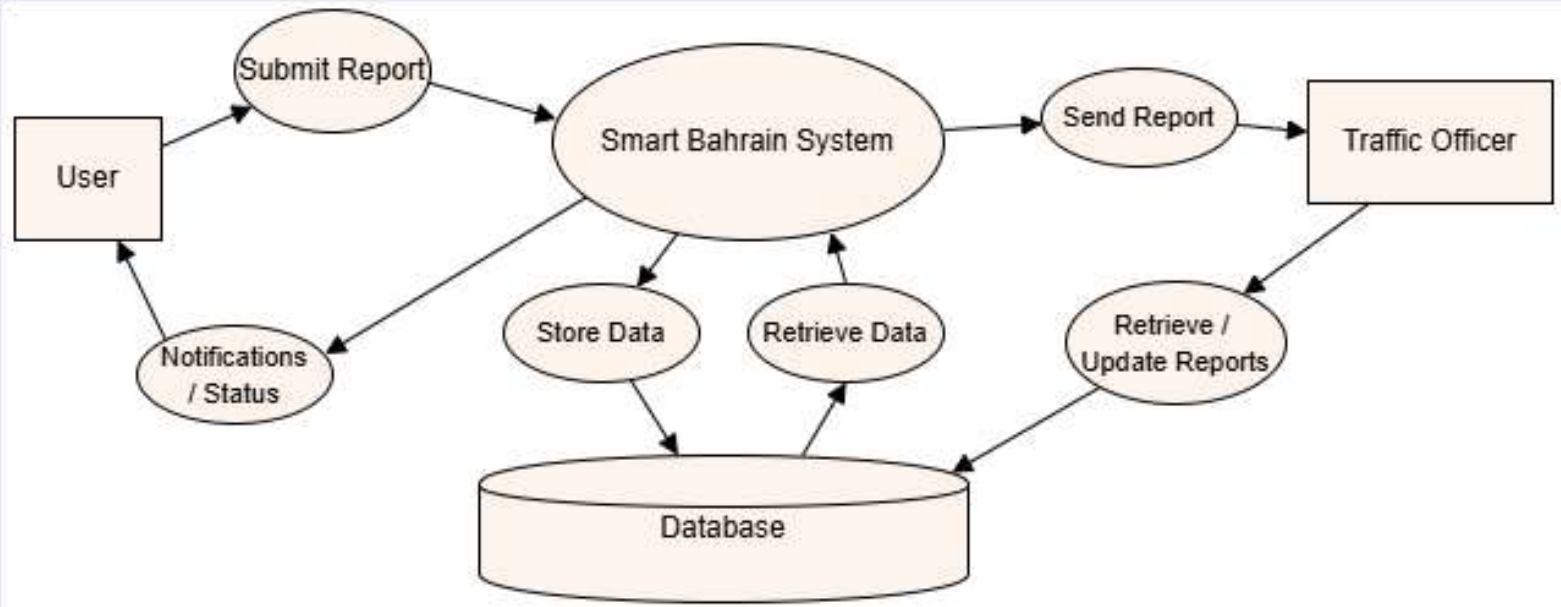
Entity Relationship Diagram



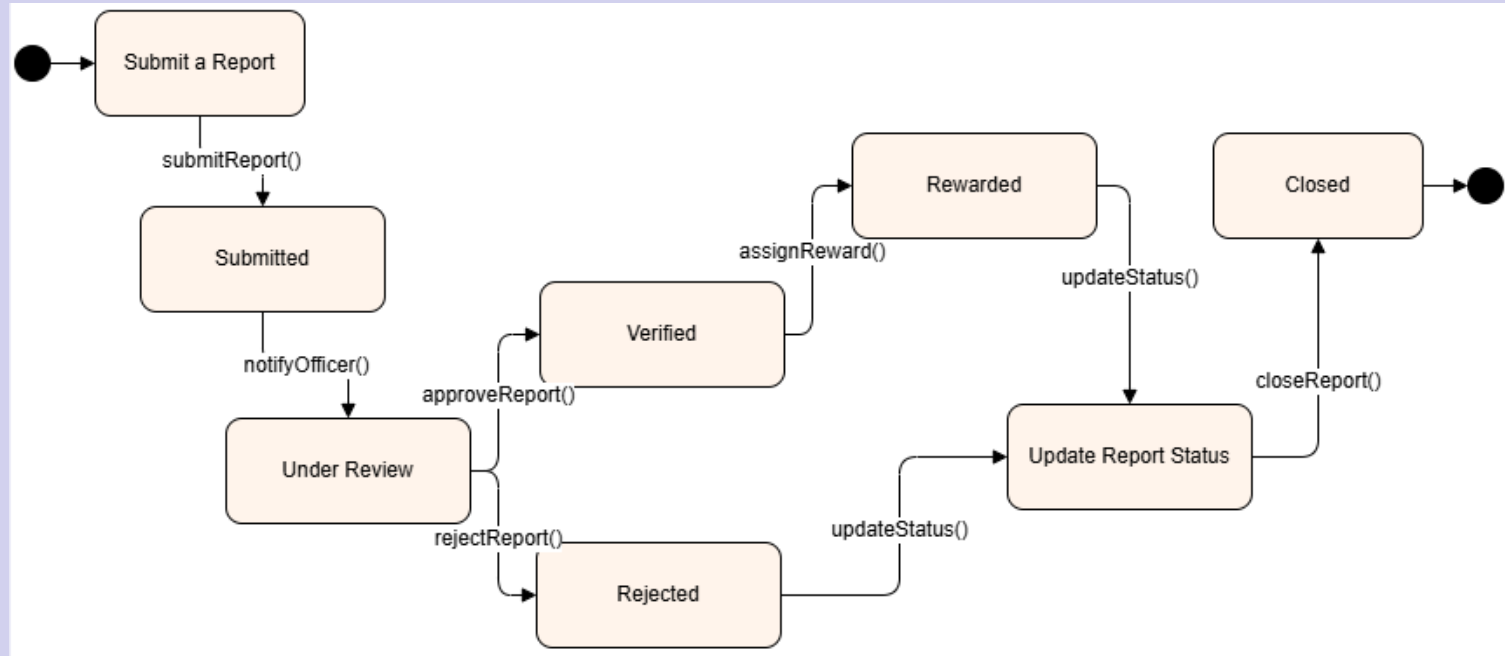
Database Diagram



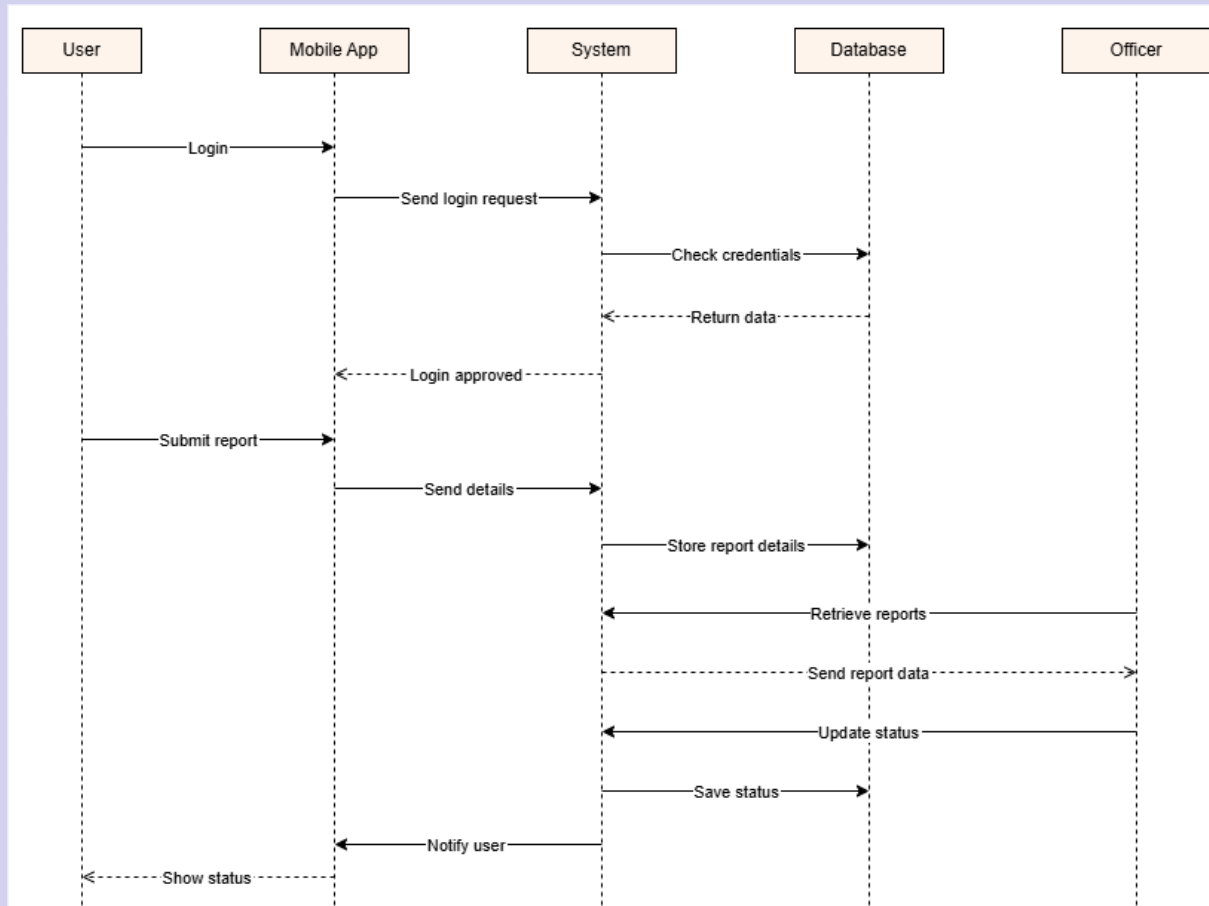
State Diagram



Data Flow Diagram



Sequence Diagram



User Interface

Report Violation

Select Type of Offense

Choose offense type

Description

Provide details about the violation

Add Photos / Videos

Tap to select media

Submit Report

Home

My Reports

Report

Rewards

Profile

3 Total Cases

1 Pending 1 Approved 1 Rejected

Submitted Reports

Illegal Parking

King Faisal Road

Verified

✓ ✕

Blocking Driveway

Adliya

Under Review

✓ ✕

Careless Driving

Seel

Rejected

✓ ✕

Home

My Reports

Report

Rewards

Profile

Rewards

Total Rewards

480

50 Points for Verified Report on Double Parking
Earned on: 01/03/2025

30 Points for Verified Report on Illegal Parking
Earned on: 12/05/2025

40 Points for Verified Report on Blocking Driveway
Earned on: 22/07/2025

Home

My Reports

Report

Rewards

Profile

THANK YOU!

